

Daily Logic Drill #6

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Mathematical Induction · Recurrence Sequences · Base-Case Traps · Strong Induction.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Induction structure, base cases, and recurrence sequences. Pure recognition.

1. A sequence is defined by $u_1 = 1$ and $u_{n+1} = 2u_n + 1$. A student claims $u_n = 2^n - 1$ for all $n \geq 1$. For which value of n does the base case verify this claim?
2. To prove $P(n)$ for all integers $n \geq 4$ by induction, what is the base-case integer?
3. If $P(k) \implies P(k+1)$ is proved for all $k \geq 1$, but no base case is ever established, has $P(n)$ been proved for any n ?
4. A sequence satisfies $u_{n+1} = u_n + 2n + 1$ with $u_1 = 1$. What is the closed-form expression for u_n ?
5. True or false: “Mathematical induction can be used to prove statements about every real number $x \geq 1$.”
6. To prove $2^n > n^2$ for all $n \geq 5$ by induction, what is the smallest integer n for which the base case must be verified?
7. What is the negation of the inductive-step statement “For all $k \geq 1$, $P(k) \implies P(k+1)$ ”?
8. True or false: “If $P(1)$ is true and $P(k) \implies P(k+2)$ holds for all $k \geq 1$, then $P(n)$ is true for every positive integer n .”
9. What is the smallest positive integer n for which $n! > 3^n$?
10. A recurrence satisfies $u_{n+2} = 3u_{n+1} - 2u_n$ with initial terms $u_1 = 1$ and $u_2 = 3$. What is the minimum number of base cases required to anchor an induction for all integers $n \geq 1$?

Section B Manipulation Drills

(≤50 s each)

Algebraic manipulations within inductive steps, and short full inductions.

1. Express $\sum_{i=1}^{k+1} i^3$ in terms of $\sum_{i=1}^k i^3$.

2. Assuming $\sum_{i=1}^k i = \frac{k(k+1)}{2}$, add $(k+1)$ and simplify to show $\sum_{i=1}^{k+1} i = \frac{(k+1)(k+2)}{2}$.
3. Prove by induction: $\sum_{i=1}^n (2i-1) = n^2$ for all positive integers n .
4. Assuming $4^{2k} - 1$ is divisible by 15, express $4^{2(k+1)} - 1$ in a form that shows it is also divisible by 15.
5. Prove by induction that $5^n - 1$ is divisible by 4 for all positive integers n .
6. Assuming $k! > 2^k$ for some $k \geq 4$, prove $(k+1)! > 2^{k+1}$.
7. Prove by induction: $2^n \geq n+1$ for all non-negative integers n .
8. Assuming $\sum_{i=1}^k \frac{1}{i(i+1)} = \frac{k}{k+1}$, show that $\sum_{i=1}^{k+1} \frac{1}{i(i+1)} = \frac{k+1}{k+2}$.
9. Prove by induction that the sum of the interior angles of a convex n -gon is $(n-2) \times 180^\circ$ for all $n \geq 3$.
10. A student wants to prove $P(n)$ for all $n \geq 1$ using base cases $P(1)$ and $P(2)$, with inductive step “ $P(k)$ and $P(k+1)$ together imply $P(k+2)$ ” for $k \geq 1$. Explain why establishing *both* $P(1)$ and $P(2)$ (not just $P(1)$) is necessary here.

Section C Substitution & Structure(≤ 90 s each)*Misapplied induction and subtle inductive-step gaps. Tricky MCQs.*

1. A student attempts to prove $P(n)$: “in any group of n people, everyone has the same favourite integer,” by induction. Base case: $P(1)$ is trivially true. Inductive step: assume $P(k)$ holds for any group of k people. Given a group of $k+1$ people, remove one person, leaving a group of k people who all share a favourite by $P(k)$; then put that person back and remove a different person, leaving another group of k people who also all share a favourite by $P(k)$. Since the two groups overlap, everyone must share the same favourite, so $P(k+1)$ holds. Which of the following identifies the flaw?
 - A) The base case $P(1)$ is invalid.
 - B) The inductive step implicitly assumes $k \geq 2$ for the subgroups to overlap.
 - C) The claim is actually true, and the proof is valid.
 - D) The flaw is that people cannot have favourite integers.
 - E) The inductive hypothesis is applied twice, which is never allowed.
2. A student attempts to prove: for all positive integers n , $n^3 - n$ is divisible by 6. Inductive step: “Assume $k^3 - k = 6m$ for some integer m . Since $(k+1)^3 - (k+1)$ is also divisible by 6, by the general theory of consecutive-integer products, $P(k+1)$ holds.” What is the flaw?
 - A) $n^3 - n$ is not actually always divisible by 6.
 - B) The inductive hypothesis is bypassed entirely.
 - C) The base case is missing.

- D) m must be shown to be positive.
- E) There is no flaw.
3. A student wants to prove $P(n)$: “ $n^2 > 3n + 4$ ” for all sufficiently large n , and sets the base case at $n = 4$, checking $16 > 16$. Since this is false, the student concludes the claim can never be proved by induction. What is the actual smallest valid base case, and what was wrong with the student’s reasoning?
- A) $n = 3$; the student made an arithmetic slip.
- B) $n = 5$; $P(4)$ is false, but $P(5)$ is true, and the inductive step holds for $k \geq 5$.
- C) $n = 6$; the student needed to check two consecutive cases before starting.
- D) There is no valid base case; the claim is false for all n .
- E) $n = 4$ actually does work; $16 > 16$ should be interpreted as $\geq$.
4. Every integer $n \geq 2$ can be written as a product of primes. A student attempts ordinary (single-step) induction: base case $n = 2$ (prime); inductive step: “assume k is a product of primes; show $k + 1$ is a product of primes.” What is the fundamental problem with this approach?
- A) There is no problem; the proof works fine.
- B) Strong induction is required because the factors of $k + 1$ may not relate to k .
- C) The base case should be $n = 1$.
- D) $k + 1$ is always prime if k is a product of primes.
- E) The inductive hypothesis needs to also assume $k - 1$ is a product of primes.
5. A student is asked to determine the smallest positive integer n_0 such that $P(n)$: “ $\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} > \frac{1}{2}$ ” holds for all $n \geq n_0$. Checking $n = 1$: the sum is just $\frac{1}{2}$, so $P(1)$ is FALSE ($\frac{1}{2} > \frac{1}{2}$ fails). The student concludes: “since $P(1)$ fails, no base case exists and induction cannot be used to establish $P(n)$ for any range of n .” Is this conclusion correct?
- A) Yes.
- B) No — the claim could still be true from some larger n_0 onward.
- C) Yes, because a failed base case always means the inductive step will also fail.
- D) No, because $\frac{1}{2} > \frac{1}{2}$ is actually a true statement.
- E) The claim is undecidable by induction in principle.
6. A student attempts to prove “every amount of postage $n \geq 8$ pence can be made using only 3p and 5p stamps” by induction, using base cases $n = 8, 9, 10$ (each directly constructible: $8 = 3 + 5$, $9 = 3 + 3 + 3$, $10 = 5 + 5$). What must the inductive step actually use to correctly bridge from an earlier case to $P(k + 1)$ for $k \geq 10$?
- A) $P(k + 1)$ relies on the construction for $P(k - 2)$, requiring strong induction.
- B) $P(k + 1)$ follows directly from $P(k)$ by adding a 1p stamp.
- C) No inductive step is needed once three base cases are given.

- D) $P(k+1)$ cannot be constructed from any earlier case.
- E) The base cases 8, 9, 10 are unnecessary.
7. A student checks that $F_1 + F_3 + F_5 + \cdots + F_{2n-1} = F_{2n}$ (where F_i is the i -th Fibonacci number, $F_1 = F_2 = 1$) for $n = 1, 2, 3, 4$ and concludes the identity holds for all positive integers n . Which of the following is the most accurate assessment?
- A) This is a complete and valid proof.
- B) Checking finitely many cases is not a proof; a full argument requires an inductive step.
- C) The identity is actually false, since Fibonacci numbers grow exponentially.
- D) No proof is possible since Fibonacci numbers are not well understood.
- E) Checking $n = 1, \dots, 4$ is sufficient.
8. Which of the following correctly and precisely states what must be proved in the inductive step of a proof that $P(n)$ holds for all integers $n \geq 1$?
- A) $P(n)$ is true for some particular large n .
- B) $P(k)$ is true for all k .
- C) For every integer $k \geq 1$: if $P(k)$ is true, then $P(k+1)$ is true.
- D) There exists an integer $k \geq 1$ such that $P(k) \implies P(k+1)$.
- E) $P(1) \implies P(n)$ for all n .

Section D Challenge

(SMC / BMO1 difficulty)

Full olympiad-style induction proofs.

1. Let x be a real number such that $t = x + \frac{1}{x}$ is an integer greater than 2. Define $t_n = x^n + \frac{1}{x^n}$ for each positive integer n (and $t_0 = 2$). (after BMO1 2015 Q4)
- (a) Show that $t_{n+1} = t \cdot t_n - t_{n-1}$ for all $n \geq 1$.
- (b) Prove by strong induction that t_n is an integer for every positive integer n .
2. A positive integer is called *charming* if it equals 2 or is of the form $3^i 5^j$ for non-negative integers i, j . Prove, using strong induction, that every positive integer can be written as a sum of distinct charming integers. (after BMO1 2016 Q6)
3. Prove by induction that for all real $x > -1$ and all positive integers n , $(1+x)^n \geq 1+nx$ (Bernoulli's Inequality). Identify precisely which step of your inductive argument requires the condition $x > -1$.
4. Let $a_1 = 2$ and $a_{n+1} = a_n^2 - a_n + 1$ for $n \geq 1$. Prove by induction that $a_1, a_2, \dots, a_n$ are pairwise coprime for every positive integer n (i.e. $\gcd(a_i, a_j) = 1$ whenever $i \neq j$).
5. Prove by induction that for all integers $n \geq 2$, $\sum_{k=1}^n \frac{1}{k^2} < 2 - \frac{1}{n}$. (You may find it helpful to bound $\frac{1}{(n+1)^2}$ against $\frac{1}{n} - \frac{1}{n+1}$ in your inductive step.)

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths